MONTH 1 Python (Foundations)

Weeks 1–2: Python Fundamentals

- Syntax overview
- Variables & data types
- Conditions & loops
- Functions
- Basic error handling
- Working with files

Weeks 3–4: Data Science Python

- **NumPy**: arrays, indexing, array operations
- **Pandas**: DataFrames, cleaning, filtering, grouping
- **Matplotlib**: basic plots, scatter, bar charts
  Small Python mini-projects

MONTH 2: Data Analysis & EDA

Weeks 1–2: Data Wrangling

- Loading different data formats
- Handling missing & dirty data
- Transformations, merging, joins

Weeks 3–4: Exploratory Data Analysis

- Descriptive summaries
- Visual EDA with Matplotlib
- Outlier detection
- Feature understanding (non-math)

MONTH 3: Machine Learning Foundations

Weeks 1–2: Supervised Learning (Regression & Classification)

- Linear Regression (intuitively)
- Logistic Regression
- Decision Trees
- Random Forests
- Evaluation metrics (non-math explanations)

Weeks 3–4: Intermediate ML

- Train/validation/test split
- Pipelines in scikit-learn
- Hyperparameter tuning (GridSearch, RandomSearch)
- Overfitting/underfitting (no math formulas)

MONTH: 4 Advanced ML & Ensembles

Weeks 1–2: Ensemble Models

- Gradient Boosting
- XGBoost
- LightGBM
- Practical model comparison

Weeks 3–4: Unsupervised Learning

- Clustering: K-Means, hierarchical
- Dimensionality reduction: PCA (conceptual)
- Anomaly detection basics

MONTH 5: NLP

Weeks 1–2: NLP Fundamentals

- Text preprocessing
- Tokenization, stopwords
- TF-IDF
- Text pipelines

Weeks 3–4: Modern NLP

- Word embeddings: Word2Vec (conceptually)
- Intro to Transformers
- BERT overview & architecture (no math)
- Fine-tuning BERT for classification (practical lab)

MONTH 6: Deep Learning

Weeks 1–2: Neural Networks Basics

- What is a neural network
- Layers, activations
- Loss functions (intuitive)
- Training a simple NN in PyTorch

Weeks 3–4: CNNs & Vision

- Convolution layers overview
- Building CNNs for image classification
- Vision Transformers (ViT) introduction
  Small computer vision project

MONTH 7: Time Series & Recommenders

Weeks 1–2: Time Series Forecasting

- Train-test splits for time series
- Moving averages, smoothing
- Practical ARIMA/Prophet usage (no math)
- Forecast evaluation

Weeks 3–4: Recommendation Systems

- Collaborative filtering
- Content-based recommenders

- Building a simple recommender system project

MONTH 8: MLOps & Deployment

Weeks 1–2: MLOps Basics

- Versioning: DVC, Git
- Experiment tracking with MLflow
- Packaging models with Docker

Weeks 3–4: Deployment on Cloud

- FastAPI model serving
- Deploying on AWS Sagemaker
- Deploying on GCP Vertex AI
- Monitoring models & drift (conceptually)

MONTH 9: Capstone Project + Career Prep

Weeks 1–3: Final Project

- Data collection & preparation
- Model building & iteration
- Deployment
- Documentation + presentation practice

Week 4: Career Readiness

- Portfolio building (GitHub, HuggingFace)
- Resume optimization for ML roles
- LinkedIn optimization
- Behavioral + technical interview prep